

Presented by Robin Sarkar

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<https://linktr.ee/robin.sarkar>

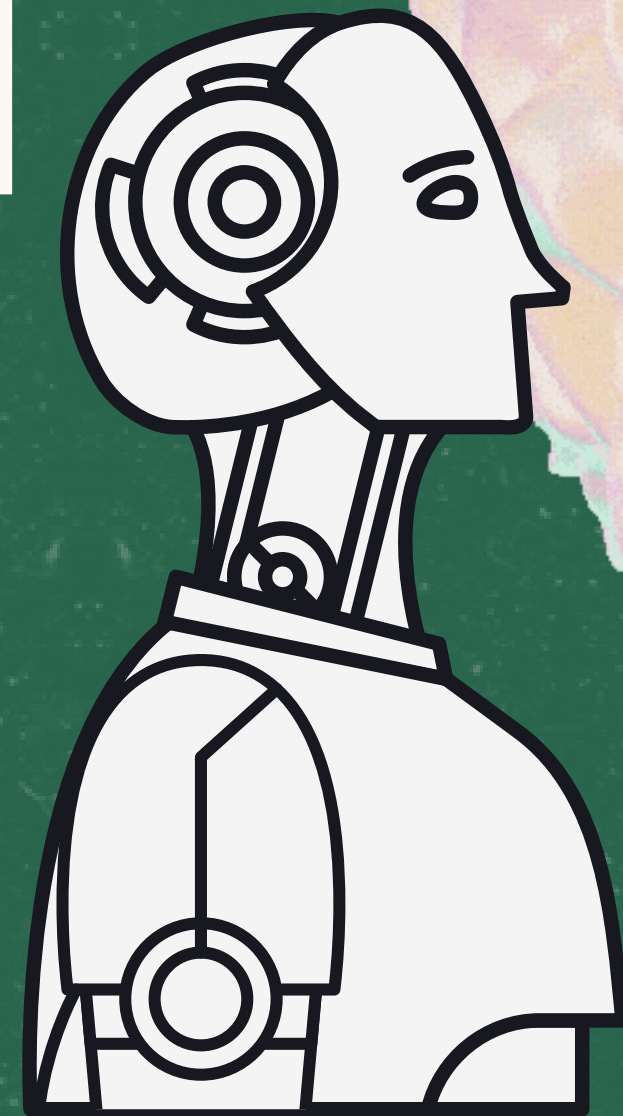
SMART TRASH BIN



ABOUT ME

My name is Robin Sarkar, and I am a student at LaGuardia Community College with a strong interest in technology and problem-solving. I enjoy exploring how simple ideas can be turned into real solutions using coding and hardware tools. Through my projects, I aim to create practical innovations that can improve everyday life and help my community. I am always eager to learn, take on challenges, and develop new skills as I work toward becoming a software engineer.

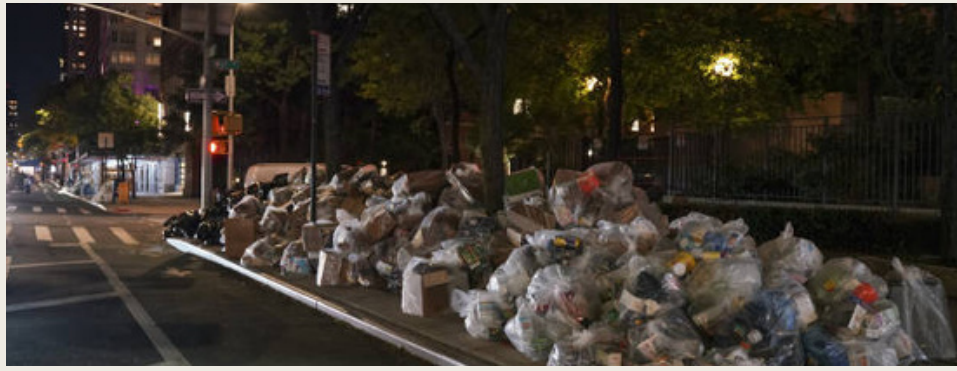
HELLO! MY NAME IS ROBIN



The Perfect City???

New York City is known for its bright lights, iconic skyline, and endless opportunities. But behind the beauty and excitement, the city faces challenges that aren't always visible. Beneath the streets and between the buildings, hidden issues like rising rat infestations reveal that even a world-class city has problems that go unnoticed. This project explores what lies behind the surface of NYC and how these hidden challenges affect everyday life.





PROBLEM



The Smart Trash Bin

The Smart Trash Bin is a simple yet efficient solution designed to prevent trash overflow in public areas. It uses an ultrasonic sensor to measure how full the bin is and an Arduino to process that data. When the bin reaches a certain level, the system sends a signal to a Raspberry Pi, which then alerts cleaning staff that the bin is almost full or completely full. This helps ensure timely waste collection, reduces overflow, and keeps public spaces cleaner. The system is low-cost, easy to install, and provides a more efficient way to manage trash in busy environments.

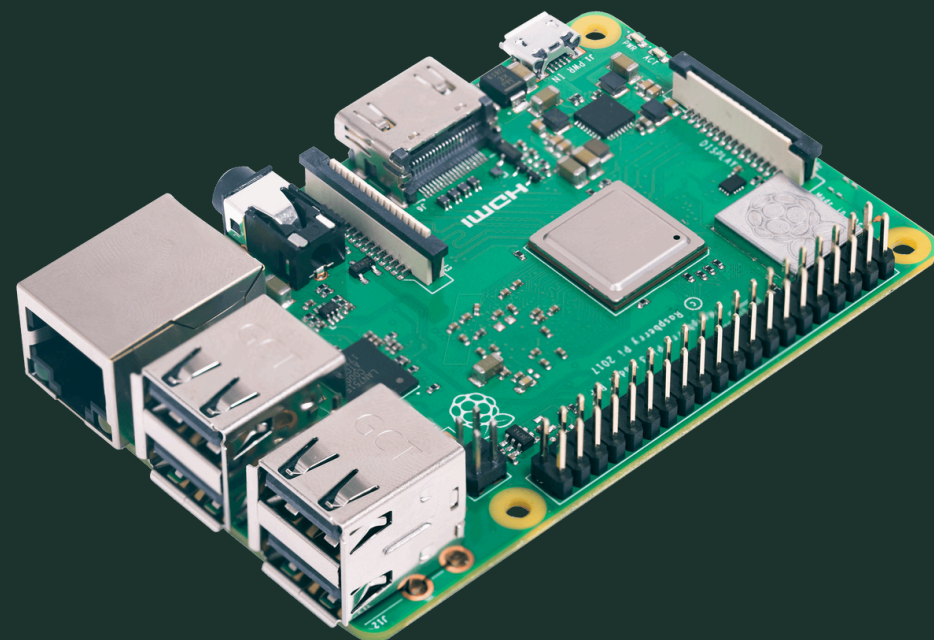


WHAT WE WILL BE NEEDING?



ARDUINO UNO

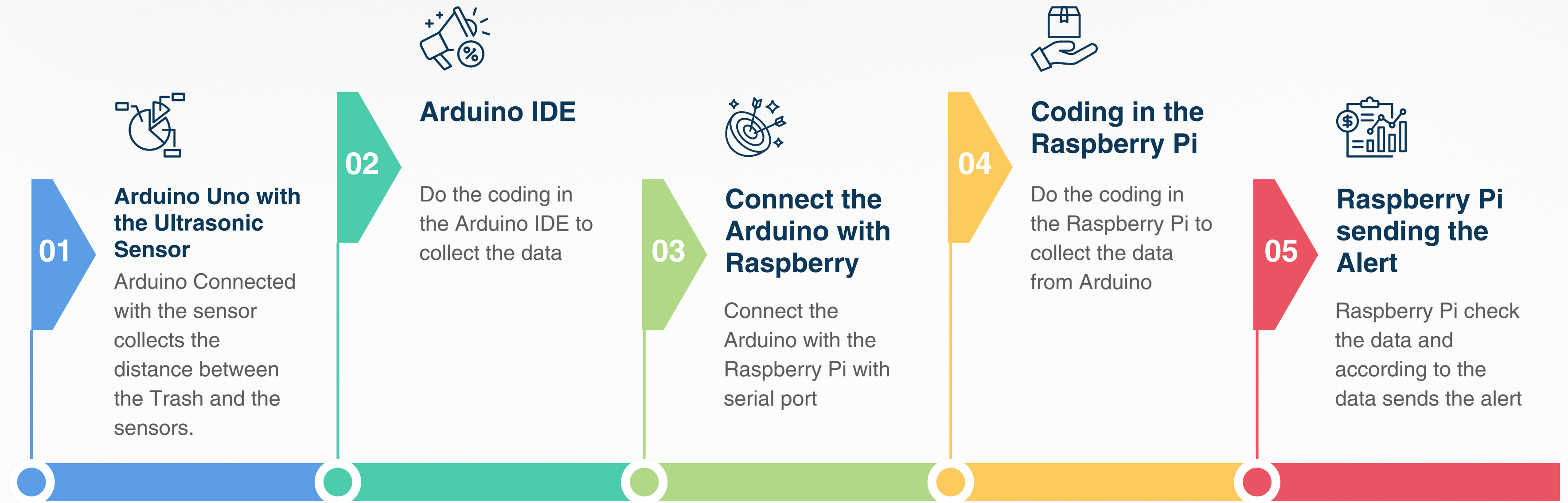
The Arduino Uno is a microcontroller board used for handling hardware tasks such as reading sensors and controlling electronic components. In the Smart Trash Bin system, the Arduino connects to an ultrasonic distance sensor that measures how full the trash bin is by detecting the distance between the trash and the sensor. It quickly processes this information and decides whether the bin is "OK," "NEAR FULL," or "FULL." Because the Arduino is designed for real-time sensor reading and simple decision-making, it is the ideal device for monitoring the trash bin's physical environment.



RASPBERRY PI

The Raspberry Pi is a microcontroller board used for handling hardware tasks such as reading sensors and controlling electronic components. In the Smart Trash Bin system, the Raspberry Pi connects to an ultrasonic distance sensor that measures how full the trash bin is by detecting the distance between the trash and the sensor. It quickly processes this information and decides whether the bin is "OK," "NEAR FULL," or "FULL." Because the Raspberry Pi is designed for real-time sensor reading and simple decision-making, it is the ideal device for monitoring the trash bin's physical environment.

PROCESS



Arduino Code: (C++)

```
float duration;  
int distance;  
int fulltrashhold = 10cm;  
int warningtrashhold = 20cm;
```

```
const int trigpin = 10;  
const int echopin = 11;  
void setup() {  
  Serial.begin(9600);  
  pinMode(trigpin, OUTPUT);  
  pinMode(echopin, INPUT);  
}
```

```
void loop() {  
  digitalWrite(trigpin, LOW);  
  delayMicroseconds(2);  
  
  digitalWrite(trigpin, HIGH);  
  delayMicroseconds(10);  
  digitalWrite(trigpin, LOW);  
  
  duration = pulseIn(echopin, HIGH);  
  distance = duration * 0.034 / 2;
```

→ Forever Loop:

```
  if distance < fulltrashhold then  
    status = "FULL"  
  Else if distance < warningtrashhold  
    status = "Near-full"  
  End if statement  
  Send status to Raspberry Pi  
  wait 2 seconds  
End Loop.
```

Raspberry Pi: (Code)

BEGIN

Connect to WiFi

Connect serial port to Arduino

Cooldown = 30 * 60 # 30 minutes.

LastSenttime = 0

Use Forever loop:

message = Read_From_Serial()

Currenttime = Get_Current_time()

if message == "FULL" then

if (CurrentTime - lastSenttime) >= cooldown then
 Send Alert ("Trash Bin is FULL, please empty it")

lastSenttime = Currenttime

End if

wait 2 seconds

Loop

BENEFITS

- PREVENTS TRASH OVERFLOW AND KEEPS PUBLIC AREAS CLEANER
- HELPS CLEANING STAFF WORK MORE EFFICIENTLY
- REDUCES UNNECESSARY TRIPS TO EMPTY HALF-FILLED BINS
- LOW-COST AND EASY TO SCALE ACROSS A CITY
- WORKS IN REAL TIME WITH AUTOMATIC NOTIFICATIONS
- IMPROVES OVERALL WASTE MANAGEMENT AND CITY HYGIENE
- REDUCES PEST PROBLEMS CAUSED BY OVERFLOWING GARBAGE

FUTURE POSSIBILITIES

- **ADD GPS TRACKING TO IDENTIFY THE EXACT LOCATION OF EACH TRASH BIN**
- **INCLUDE SOLAR PANELS TO POWER THE SYSTEM SUSTAINABLY**
- **USE A MOBILE APP FOR REAL-TIME ALERTS AND MONITORING**
- **ADD DATA ANALYTICS TO TRACK FILL PATTERNS OVER TIME**
- **IMPLEMENT A CAMERA MODULE FOR DETECTING ILLEGAL DUMPING**
- **USE MACHINE LEARNING TO PREDICT WHEN THE BIN WILL BE FULL**
- **ADD A BATTERY BACKUP FOR UNINTERRUPTED OPERATION**

